

1. QA Test Execution Summary

This report provides formal quality assurance execution evidence for Sprint 2 test specifications covering Validation (VAL), Integration (INT), Authentication (AT), Authorization (AZ), and Security/Privacy (ST). All 20 specified test cases were executed via automated Jest and Supertest test scripts against the system backend with a 100% pass rate.

2. Complete Test Cases & Execution Matrix

Test ID	Category	US #	Test Feature	Test Condition & Steps	Expected Result	Status
VAL-S2-07	Validation / QA	US-15	Self-Request Block	Owner attempts to join own published ride offer.	Self-request is rejected with HTTP 422.	PASSED
VAL-S2-08	Validation / QA	US-15	Unauthenticated Join	Unauthenticated user attempts to submit join request.	Authentication required; rejected with HTTP 401.	PASSED
INT-S2-01	Integration / QA	US-13	Search Backend Match	Search active offers by route and compare with DB records.	Search results match eligible backend offer records exactly.	PASSED
INT-S2-02	Integration / QA	US-14	Offer Details Sync	Fetch public offer details for an active ride offer.	Latest stored offer route, time, and seat data displayed.	PASSED
INT-S2-03	Integration / QA	US-15	Request DB Persistence	Submit valid join request and inspect DB document.	Correct Pending JoinRequest record persisted in MongoDB.	PASSED
INT-S2-04	Integration / QA	US-16	Own Request Retrieval	Authenticated requester queries /api/join-requests/mine.	Only correct own request records/ statuses returned.	PASSED
INT-S2-05	Integration / QA	US-15	Server Validation	Submit duplicate or invalid join request payload to API.	Server blocks invalid creation with appropriate status code.	PASSED
INT-S2-06	Integration / QA	US-13	Search Availability Sync	Offer marked inactive or full; search repeated.	Full or inactive offer is excluded from search results.	PASSED
INT-S2-07	Integration / QA	US-13-16	Token Authorization	Omit Authorization header when calling protected endpoints.	API rejects request as unauthorized (HTTP 401).	PASSED
AT-S2-01	Auth / QA	US-13-16	Valid User Access	Authenticated user accesses protected endpoints.	Authenticated user accesses permitted functions correctly.	PASSED
AT-S2-02	Auth / QA	US-13-16	No Session Block	Access protected endpoint without active login session.	Access blocked with HTTP 401.	PASSED
AT-S2-03	Auth / QA	US-15	Join Without Login	Submit join request without authentication token.	Request rejected until valid login token is provided.	PASSED
AT-S2-04	Auth / QA	US-16	Invalid/Expired Token	Access request status using expired or invalid token.	Login required; unauthorized response returned.	PASSED
AZ-S2-01	AuthZ / QA	US-13-16	Unauthorized Action	Unauthorized user attempts action on another's offer.	Unauthorized action is blocked with HTTP 403.	PASSED
AZ-S2-02	AuthZ / QA	US-16	Cross-User Request Privacy	User attempts to inspect another user's request details.	Access is denied with HTTP 403.	PASSED
AZ-S2-03	AuthZ / QA	US-16	Permitted Request View	Requester retrieves their own join request details.	Own permitted request is accessible and returned.	PASSED
AZ-S2-04	AuthZ / QA	US-15	Ownership Self-Request	Offer owner attempts to join own ride offer.	Action is blocked by server-side ownership validation.	PASSED
AZ-S2-05	AuthZ / QA	US-16	Direct API Tampering	User tampers with request ID parameter to decide another's req.	Access is denied with HTTP 403 Forbidden.	PASSED
ST-S2-01	Security / QA	US-13-16	Protected Endpoints	Access protected API routes without Authorization header.	Blocked with HTTP 401 Unauthorized across all endpoints.	PASSED
ST-S2-02	Security / QA	US-16	Privacy Data Exposure	Non-owner inspects public offer endpoint.	Private driver/requester contact details not exposed.	PASSED

3. Automated Execution Output

Automated execution performed via Jest test runner: 20/20 test cases in sprint2-qa-matrix.test.js passed cleanly. Total server test matrix across all sprints stands at 96 passed tests across 7 test suites (Sprint 1, Sprint 2 TDD/UT, Sprint 2 QA, Sprint 3 TDD/UT, Sprint 3 AUTO, Validators, Auth).

